

Beyond Accuracy

Selecting the Right Cardiovascular Disease Classifier for Clinical Referral Decisions

BU MET CS 577 A2 | Data Science with Python | Spring 2026

Professor: Eugene Pinsky | Final Project

Abstract. Primary care doctors decide every day whether to refer a patient to a cardiologist or send them home. Most machine learning projects answer this by picking the classifier with the highest accuracy. This project shows that is the wrong objective. A false negative (sending a sick patient home) costs roughly \$150,000 in emergency care. A false positive (unnecessary referral) costs \$300. That is a 500 to 1 cost asymmetry. When six classifiers are reranked by actual clinical cost instead of accuracy, the leaderboard flips completely. KNN, which ranked last on accuracy, turns out to be the safest clinical choice. This conclusion holds across every cost assumption tested from \$25,000 to \$200,000. The project also shows that just three measurements (blood pressure, cholesterol, and age) capture 99.5% of full model accuracy, all six models pass a gender fairness audit, and five of six are robust to 20% measurement noise.

1. Introduction

Primary care doctors see patients for routine checkups every day. When a patient shows signs of potential cardiovascular disease (CVD), the doctor faces a referral decision: send them to a cardiologist or send them home. This decision is made with limited time, usually around 15 minutes per patient, and limited information.

The standard approach in machine learning is to train several classifiers, compare their accuracy, and pick the winner. This project argues that accuracy is the wrong metric for this problem. The reason is a severe cost asymmetry between the two types of mistakes a classifier can make.

False Negative (FN): the model predicts no CVD but the patient has it. The patient is sent home. Without treatment, this can lead to a heart attack, emergency hospitalization, surgery, and recovery. The estimated total economic cost is around \$150,000.

False Positive (FP): the model predicts CVD but the patient is healthy. The patient gets an unnecessary cardiology referral. Cost is around \$300.

That is a 500 to 1 cost ratio. Accuracy treats both mistakes as equal. In clinical reality they are not even close. This project builds a framework that accounts for that asymmetry and asks: which model saves the most lives and money when you count the actual cost of being wrong?

2. Dataset

The dataset used is the Cardiovascular Disease Dataset by Sulianova, available on Kaggle. It contains 70,000 patient records with 13 columns including lifestyle factors, physical measurements, and a binary target variable indicating whether the patient has cardiovascular disease.

Detail	Value
Source	Kaggle (Sulianova)
File	cardio_train.csv
Separator	Semicolon (;) not comma
Rows	70,000
Columns	13
Target	cardio (0 = No CVD, 1 = Has CVD)
Class balance	50.0% / 50.0%
Missing values	0

The 13 features used for modeling are:

Feature	Type	Description
age	Continuous	Age in years (originally stored in days, divided by 365.25)
gender	Binary	1 = Female, 2 = Male
height	Continuous	Height in centimeters

weight	Continuous	Weight in kilograms
ap_hi	Continuous	Systolic blood pressure (mmHg) - pressure when heart contracts
ap_lo	Continuous	Diastolic blood pressure (mmHg) - pressure when heart relaxes
cholesterol	Ordinal	1 = Normal, 2 = Above Normal, 3 = Well Above Normal
gluc	Ordinal	Blood glucose: 1 = Normal, 2 = Above Normal, 3 = Well Above Normal
smoke	Binary	0 = Non-smoker, 1 = Smoker
alco	Binary	0 = No alcohol, 1 = Drinks alcohol
active	Binary	0 = Not active, 1 = Physically active
bmi	Engineered	Body Mass Index = $\text{weight} / (\text{height in m})^2$
pulse_pressure	Engineered	ap_hi minus ap_lo. Measures arterial stiffness

Note: bmi and pulse_pressure (highlighted green) were engineered by us in Phase 2. They were not in the original dataset.

3. Data Loading and First Inspection

Before any cleaning or modeling, we ran 8 checks on the raw data. The goal was to understand what problems existed before deciding how to fix them.

Shape: 70,000 rows x 13 columns

Columns: confirmed all 13 present, id column identified for removal

Data types: age stored as integer (days, not years), all others numeric

Missing values: zero missing values across all columns

Duplicates: 0 found at this stage (24 found after dropping id in Phase 2)

Target balance: exactly 50.0% No CVD and 50.0% Has CVD

Descriptive statistics: ap_hi max = 16,020 mmHg (impossible), ap_lo min = -70 (impossible)

Red flags: age in days, extreme BP values, height of 250cm, weight of 10kg

Phase 1 - Raw Data Snapshot (Before Cleaning)

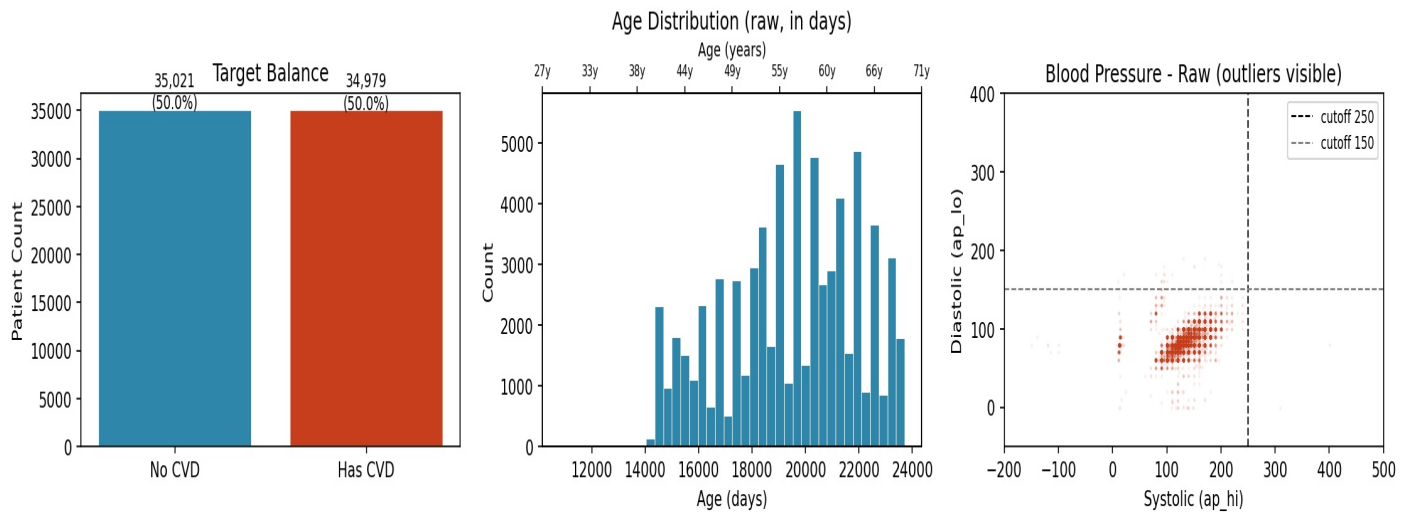


Figure 1. Raw data snapshot showing target balance, age distribution in days, and blood pressure outliers before any cleaning.

4. Data Cleaning

We applied 7 cleaning steps in sequence. Each step was logged with the exact number of rows removed so the process is fully transparent.

Step 1: Drop ID column

The id column is a row counter with no clinical meaning. It was dropped before duplicate checking because two rows with different ids but identical clinical values would not be flagged as duplicates otherwise.

Step 2: Remove duplicates

After dropping id, 24 duplicate rows were found and removed. These were not visible in Phase 1 because each row had a unique id. Keeping duplicates risks a patient appearing in both train and test sets, which would artificially inflate accuracy.

Step 3: Convert age from days to years

Age was stored in days (range: 10,798 to 23,713). We divided by 365.25 (not 365) to account for leap years. Over a 65-year lifespan the difference is about 16 days.

Step 4: Blood pressure 3-layer filter

Layer 1: systolic (ap_hi) must be between 70 and 250 mmHg. Removed 229 rows. Layer 2: diastolic (ap_lo) must be between 40 and 150 mmHg. Removed 1,017 rows (the largest single filter). Layer 3: systolic must be strictly greater than diastolic. A reading where diastolic equals or exceeds systolic is physically impossible because systolic is the peak pressure when the heart contracts and must always be higher. Removed 88 rows that passed both range checks individually but failed this logic check.

Step 5: Height and weight filters

Height: 100 to 220 cm. Removed 27 rows. Weight: 30 to 200 kg. Removed 6 rows. Our height floor of 100cm is more inclusive than common alternatives like 120cm, which would cut real short adult patients.

Step 6: Engineer BMI and cap at 60

$BMI = weight / (height \text{ in meters})^2$. We capped BMI at 60 to remove 34 rows where a patient passed height and weight filters individually but produced a physically impossible BMI when combined (e.g., 100cm height and 200kg weight gives BMI of 200). We also created bmi_category using WHO bins for visualization.

Step 7: Engineer pulse pressure

Pulse pressure = ap_hi minus ap_lo. This measures arterial stiffness and is an independent cardiovascular risk factor. Normal pulse pressure is 40 to 60 mmHg. Above 60 indicates arteries that are not flexible enough to absorb the pressure wave between heartbeats.

Metric	Value
Original rows	70,000
Rows removed	1,425 (2.0%)
Clean rows	68,575
Target balance after cleaning	50.5% No CVD / 49.5% Has CVD
Largest single filter	Diastolic BP range: 1,017 rows

Phase 2 - Cleaning Results

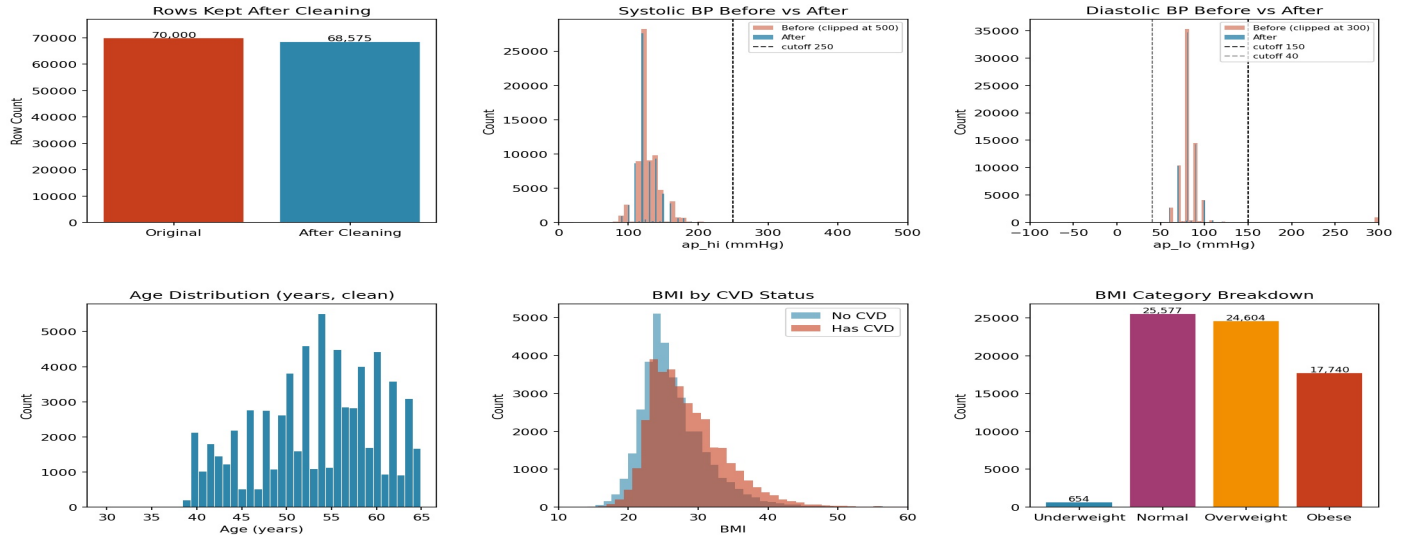


Figure 2. Cleaning results: rows kept, systolic and diastolic BP before/after, age distribution, BMI analysis.

5. Exploratory Data Analysis

EDA was structured around a single question for each figure: what does this tell us about the relationship between patient features and CVD risk? We produced 8 figures covering age, blood pressure, BMI, categorical features, correlations, gender distribution, and pairwise relationships.

Age Analysis

CVD patients have a median age of 55.8 years versus 52.0 years for healthy patients, a difference of 3.8 years. While this gap is statistically significant across 68,000 patients, the distributions overlap substantially, meaning age alone cannot separate the two groups. The box plot shows both groups span the same range (roughly 30 to 65 years).

Fig 3 - Age vs CVD Status

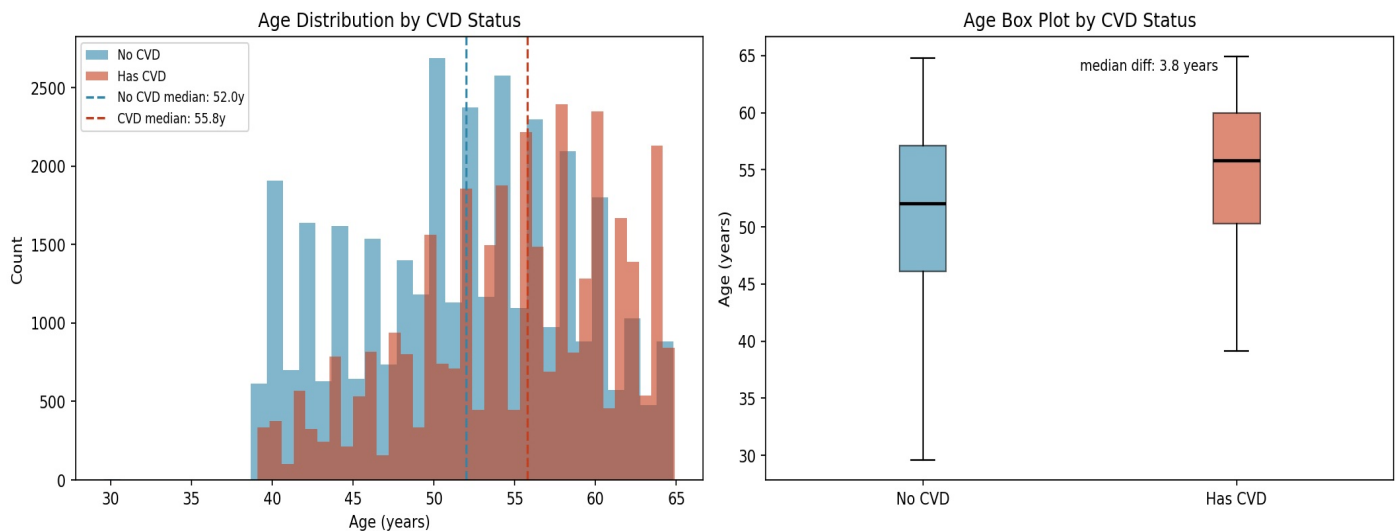


Figure 3. Age distribution and box plot by CVD status. Median difference is 3.8 years.

Blood Pressure Analysis

Systolic BP shows the clearest separation of any single feature. CVD patients have a mean systolic of 134 mmHg versus 120 mmHg for healthy patients, a 14 mmHg difference. The 134 mmHg mean sits right at the American Heart Association's hypertension stage 1 threshold. The pulse pressure box plot confirms that CVD patients have higher arterial stiffness, with median pulse pressure above 50 mmHg versus around 40 mmHg for healthy patients.

Fig 4 - Blood Pressure vs CVD Status

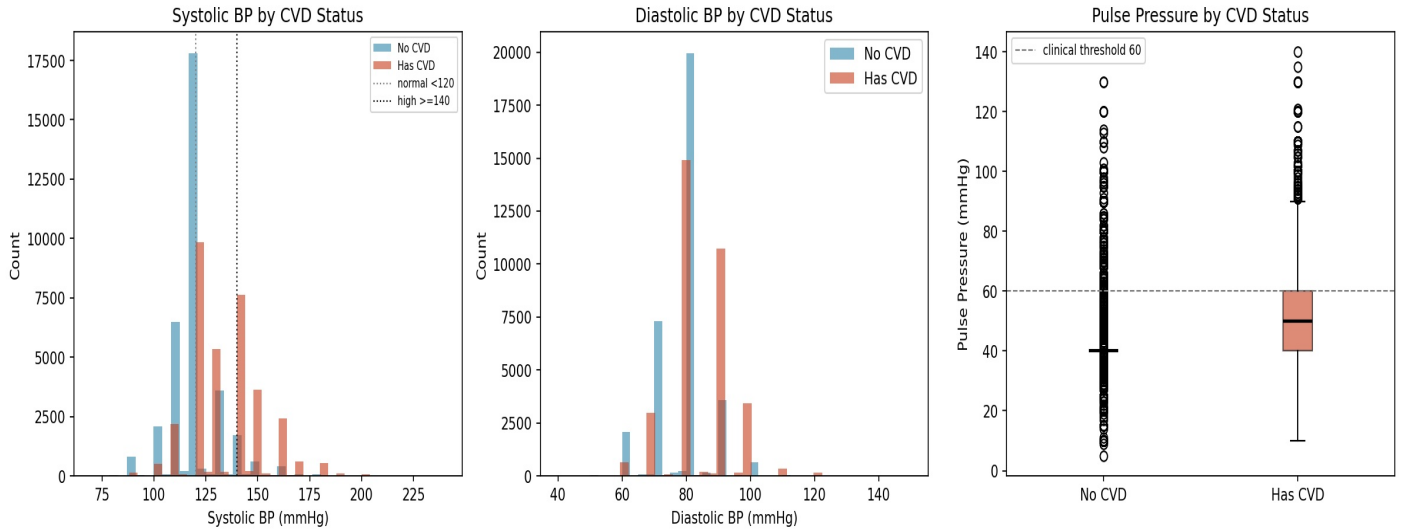


Figure 4. Systolic BP, diastolic BP, and pulse pressure by CVD status.

BMI Analysis

BMI shows a clear dose-response relationship with CVD risk. Underweight patients have a 28.0% CVD rate. Normal BMI patients have 39.8%. Overweight patients reach 50.7%. Obese patients reach 62.5%. Every category step up increases CVD rate, which is consistent with established clinical literature on obesity as a cardiovascular risk factor.

Fig 5 - BMI vs CVD Status

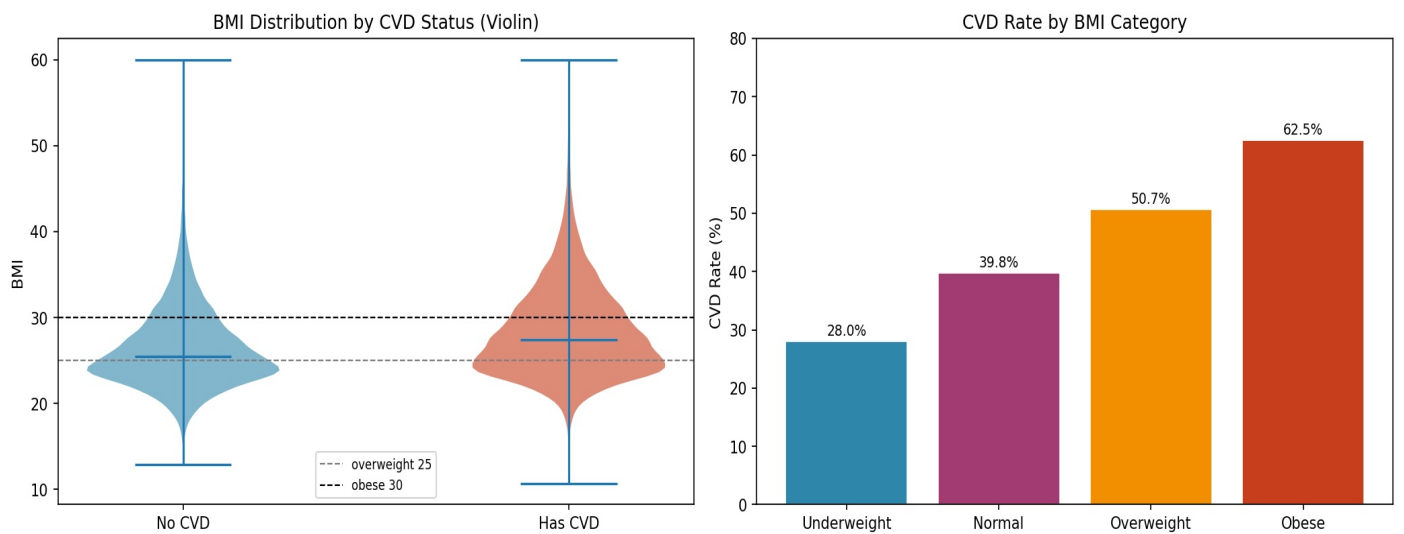


Figure 5. BMI violin plots by CVD status and CVD rate by WHO BMI category.

Categorical Features

Cholesterol shows the strongest signal among categorical features: 44% CVD rate at normal levels rising to 76% at well above normal, a 32 percentage point jump. Glucose shows a moderate effect (48% to 62%). Smoking and alcohol show a counterintuitive result: smokers and drinkers have slightly lower CVD rates than non-smokers and non-drinkers. This is a known artifact of self-reported lifestyle data. Patients may underreport these behaviors in a medical setting, making the feature unreliable as a predictor.

Fig 6 - Categorical Features vs CVD Rate

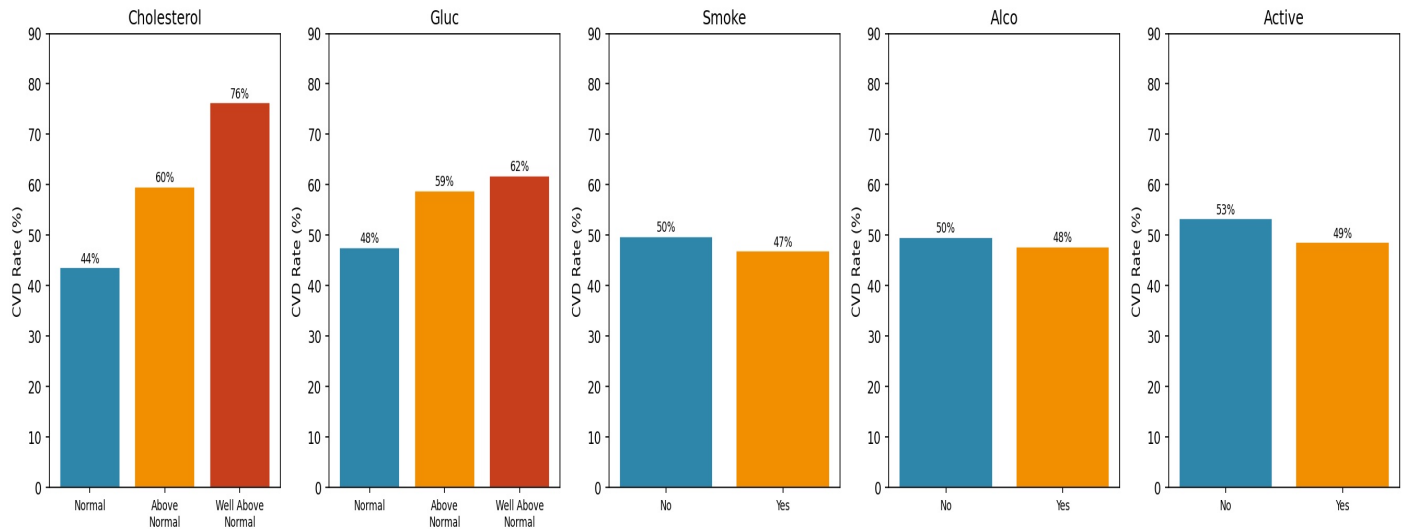


Figure 6. CVD rate per category for all five categorical features.

Correlation Analysis

The correlation heatmap shows that ap_hi has the strongest linear correlation with the target at 0.43. Cholesterol follows at 0.22. Age is at 0.24. The high correlation between bmi and weight (0.86) and between pulse_pressure and ap_hi (0.84) reflects that these features are mathematically derived from each other. For tree-based models this creates no issue. For Logistic Regression it can inflate coefficient variance, which we account for in the interpretation of LR results.

Fig 7 - Correlation Heatmap (All Features)

Lower Triangle Only (avoids redundancy)

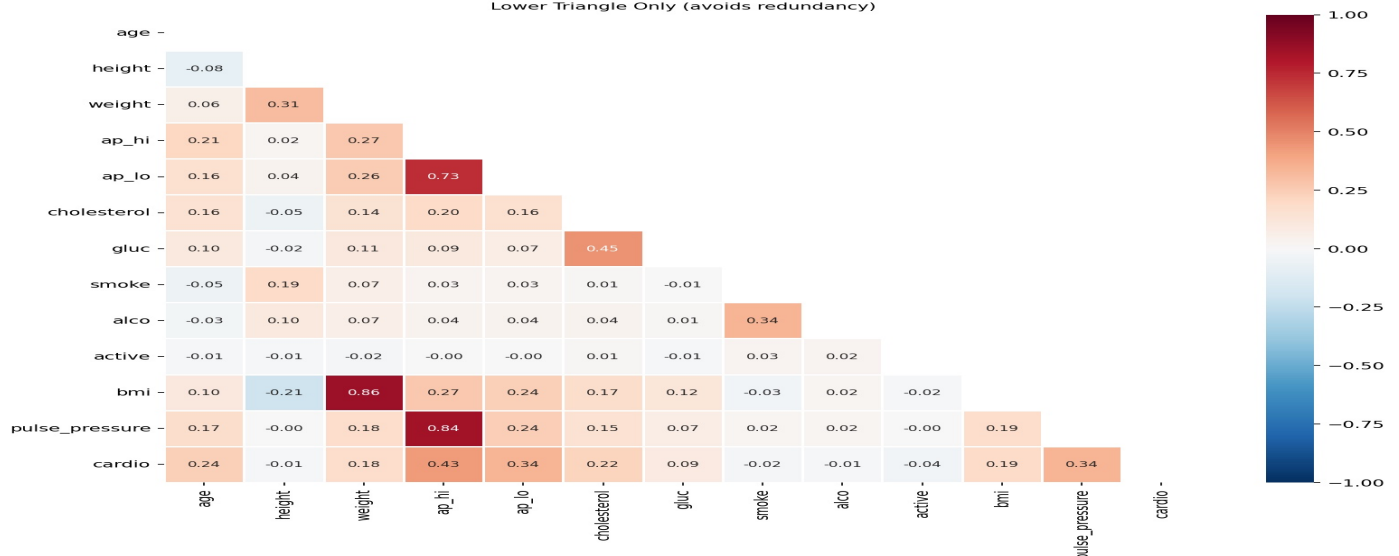


Figure 7. Correlation heatmap (lower triangle). ap_hi shows strongest correlation with CVD target at 0.43.

Age and CVD Rate by Group

Breaking age into groups reveals how sharply CVD risk rises with age. Patients aged 30 to 35 have 0% CVD rate (though the sample is only 4 patients). By ages 36 to 40, the rate is 24%. By 41 to 45 it is 31%. By 51 to 55 it reaches 46%. The 50% threshold is crossed between 55 and 60, where CVD rate reaches 56%. Patients over 60 have a 67% CVD rate.

Fig 10 - CVD Rate Rises Sharply With Age

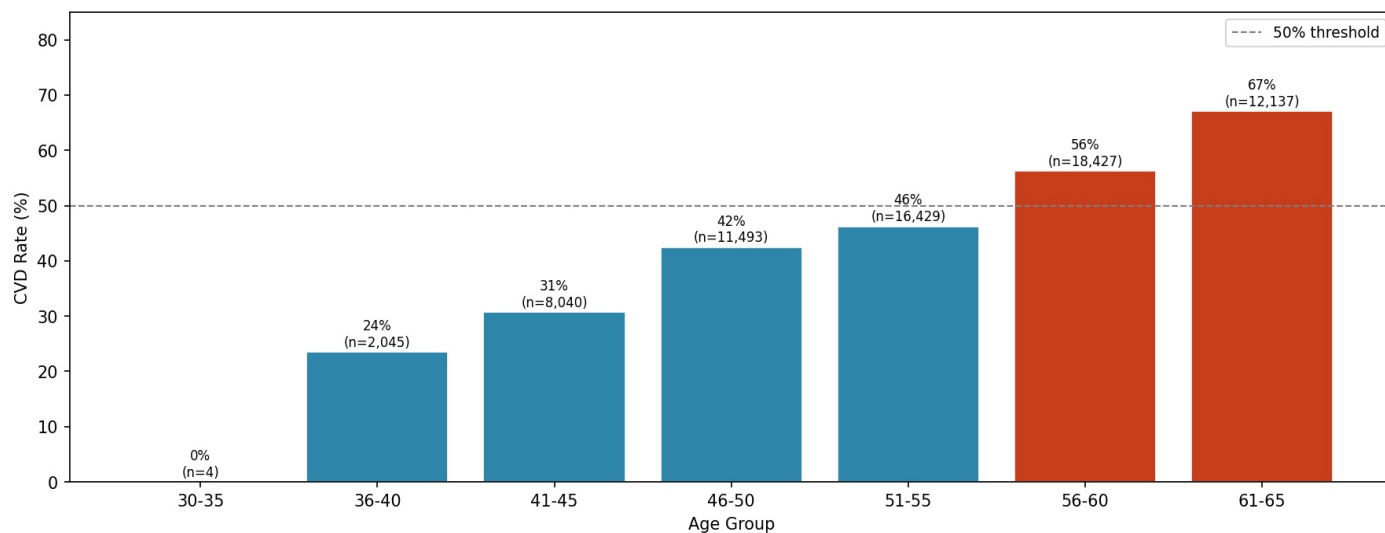


Figure 10. CVD rate by age group. Bars turn red at the 50% threshold, showing where CVD becomes the majority outcome.

6. Feature Preparation

Feature preparation followed three steps: define features and target, split into train and test sets, and scale continuous features.

Train/Test Split

We used a 70/30 stratified split with random state 42. With 68,575 rows, this gives 48,002 training rows and 20,573 test rows. Stratified splitting ensures both sets maintain the same class balance. After splitting, the CVD rate in both train and test was exactly 49.5%, confirming successful stratification.

Feature Scaling

We applied StandardScaler to the 7 continuous columns: age, height, weight, ap_hi, ap_lo, bmi, and pulse_pressure. The scaler was fitted on the training set only and then used to transform both train and test. Fitting on the full dataset before splitting would constitute data leakage because the scaler would learn statistics from the test set, which the model is not supposed to have seen during training. The 6 binary and ordinal columns (gender, cholesterol, gluc, smoke, alco, active) were left unscaled.

SVM Subsample

SVM with RBF kernel has training complexity of $O(n^2)$ to $O(n^3)$. Training on all 48,002 rows would take 10 to 30 minutes. We drew a stratified random sample of 10,000 rows for SVM training. The CVD rate in the subsample was 49.7%, confirming the sample preserved class balance. All other models trained on the full 48,002 rows.

7. Classifier Showdown

We trained 6 classifiers covering all 6 course modules. Each model used 10-fold stratified cross validation for stability estimation and was evaluated on the held-out test set of 20,573 patients.

Model	Module	Key Hyperparameters
Logistic Regression	3	max_iter=1000, random_state=42
KNN	3	n_neighbors=11 (odd to avoid ties)
Naive Bayes	5	GaussianNB, no hyperparameters
Decision Tree	5	max_depth=8, random_state=42
Random Forest	5	n_estimators=100, max_depth=12, n_jobs=-1
SVM (RBF)	6	kernel=rbf, probability=True, 10K subsample

Results

Model	Accuracy	F1	Recall	AUC	CV Mean	CV Std	Time
Random Forest	0.734	0.718	0.684	0.801	0.733	0.003	0.3s
Logistic Regression	0.729	0.711	0.672	0.793	0.727	0.004	0.0s
Decision Tree	0.730	0.714	0.681	0.790	0.727	0.004	0.1s
SVM (RBF)	0.730	0.711	0.670	0.789	0.732	0.011	4.7s
Naive Bayes	0.716	0.681	0.613	0.781	0.713	0.005	0.0s
KNN	0.719	0.711	0.697	0.774	0.718	0.004	0.0s

Orange row = accuracy winner. Green row = clinical cost winner (see Section 8).

Fig 12 - Model Comparison Across Metrics

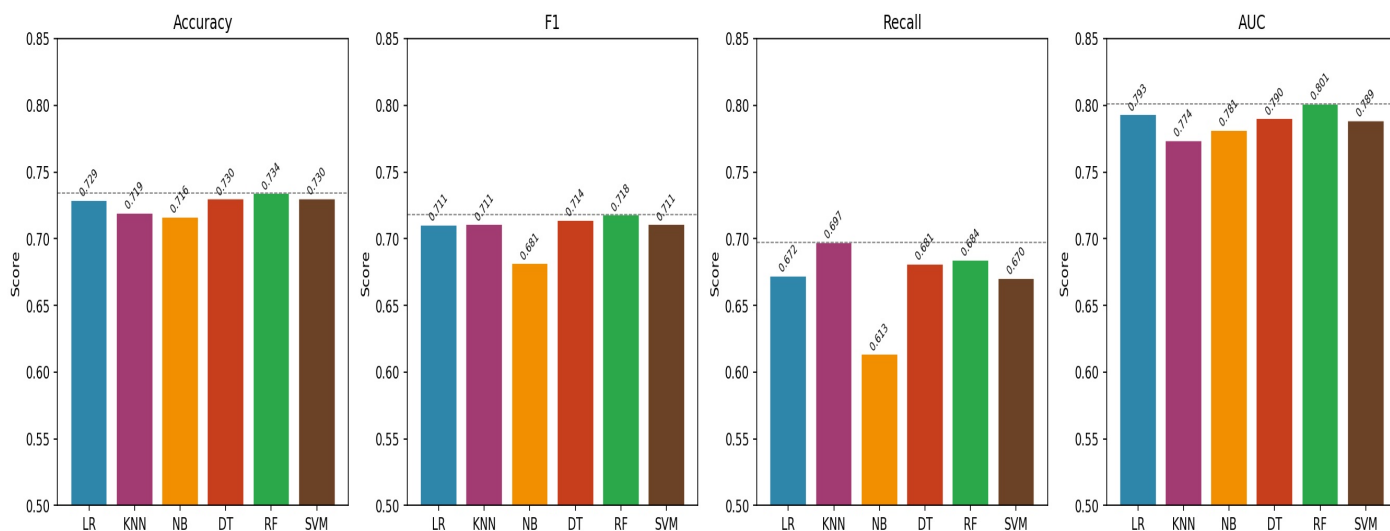


Figure 12. Model comparison across accuracy, F1, recall, and AUC.

Fig 14 - ROC Curves (All 6 Models)

Higher AUC = better discrimination

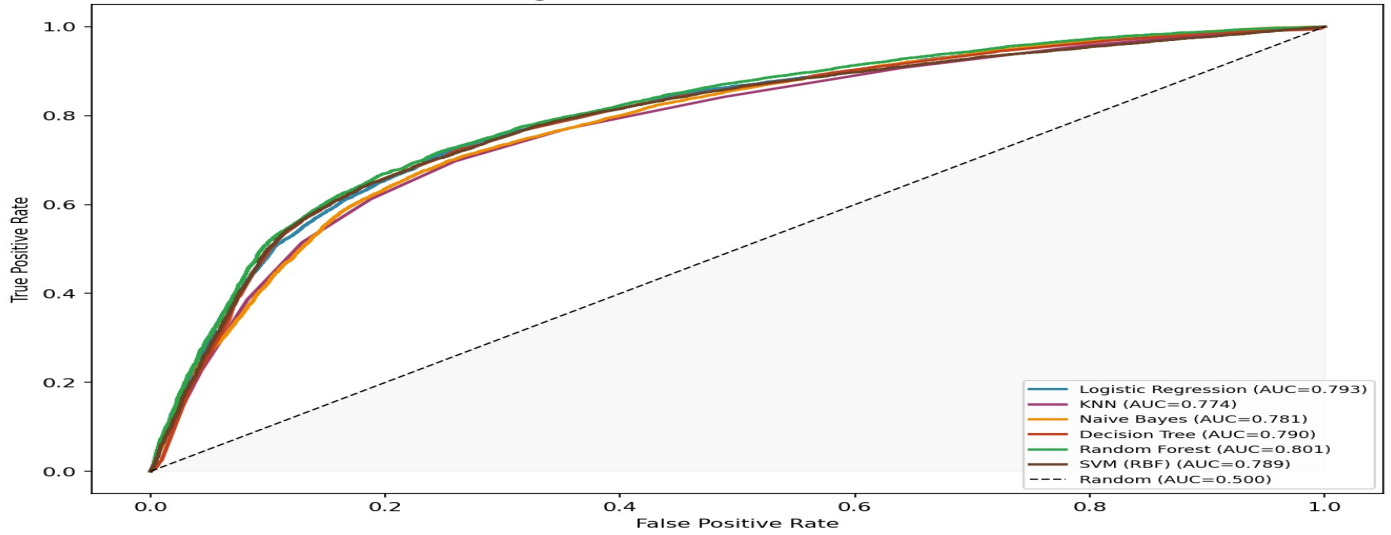


Figure 14. ROC curves for all 6 models. All curves cluster between AUC 0.774 and 0.801, indicating no single model dramatically outperforms the others on raw discrimination.

Fig 13 - 10-Fold Cross-Validation Accuracy Distribution

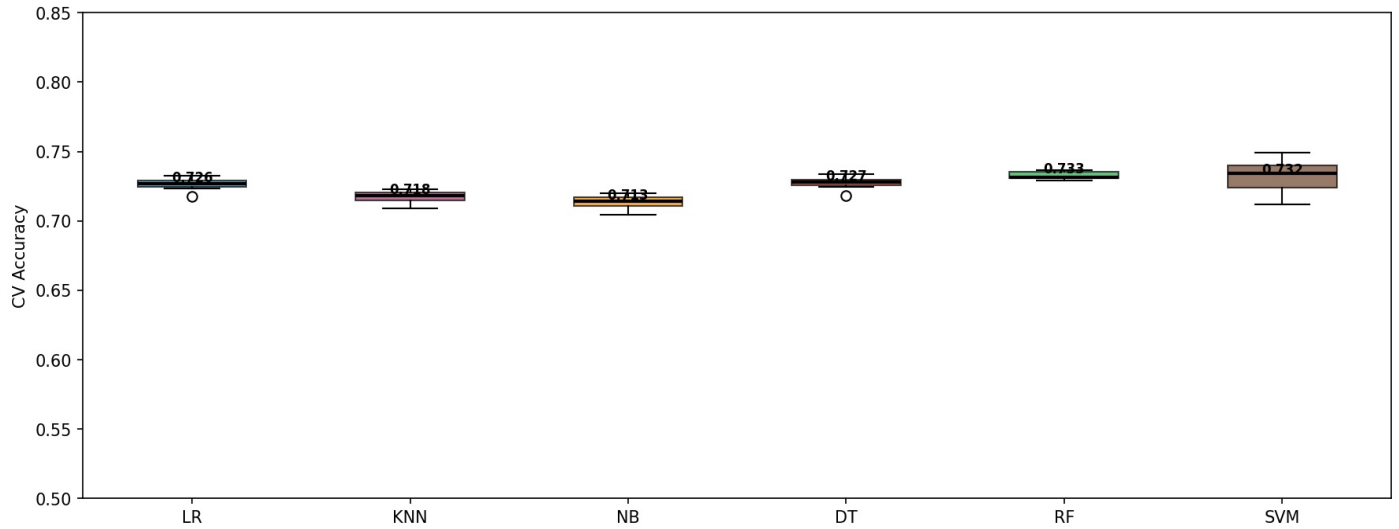


Figure 13. 10-fold CV accuracy distribution. Random Forest shows the tightest distribution (std 0.003). SVM shows the widest (std 0.011) due to the 10K subsample constraint.

Fig 15 - Confusion Matrices (All 6 Models)

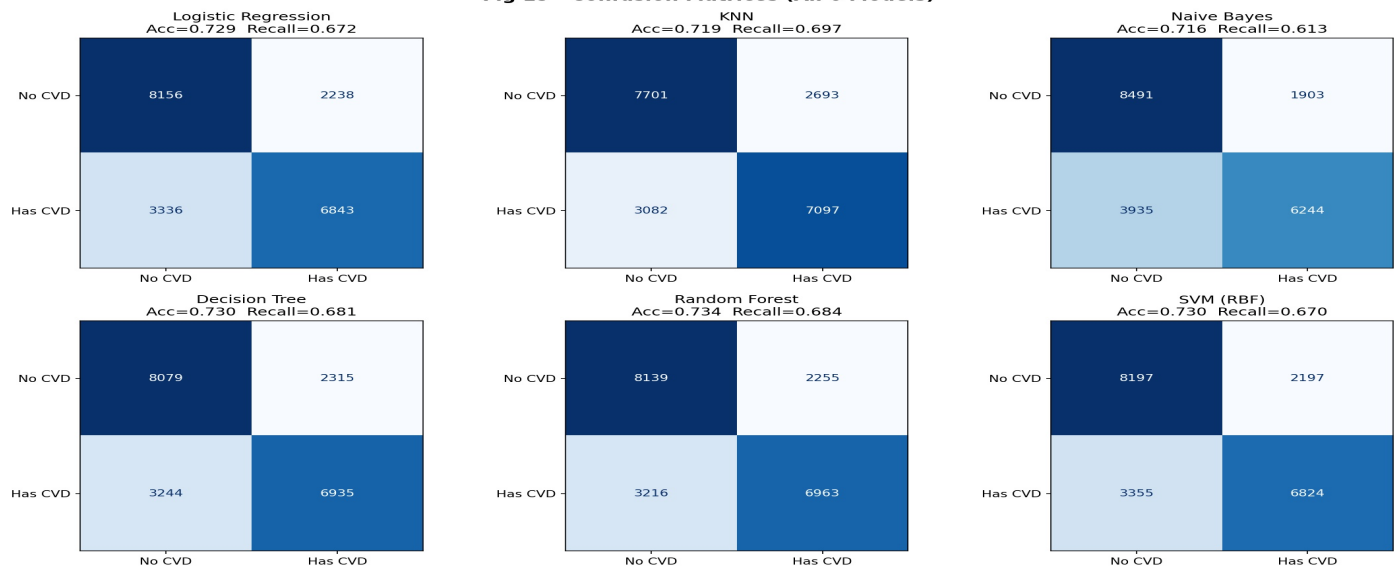


Figure 15. Confusion matrices for all 6 models. The bottom-left cell (false negatives) is the most important clinically.

7.1 Visualization Analysis

Fig 16 - Decision Boundaries (PCA 2D Projection)

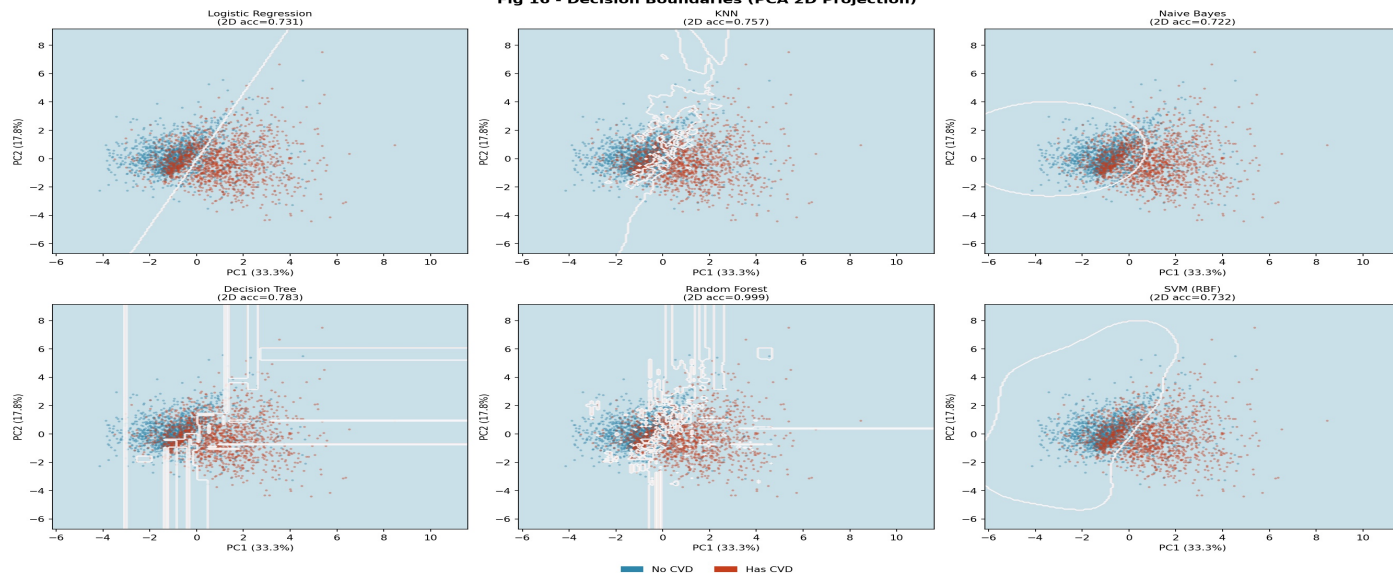


Figure 16. Decision boundaries via PCA 2D projection. LR draws a linear boundary. DT shows rectangular regions. RF smooths those regions. SVM draws a curved margin. KNN creates irregular local boundaries.

Fig 17 - KNN Optimal K (Bias-Variance Tradeoff)

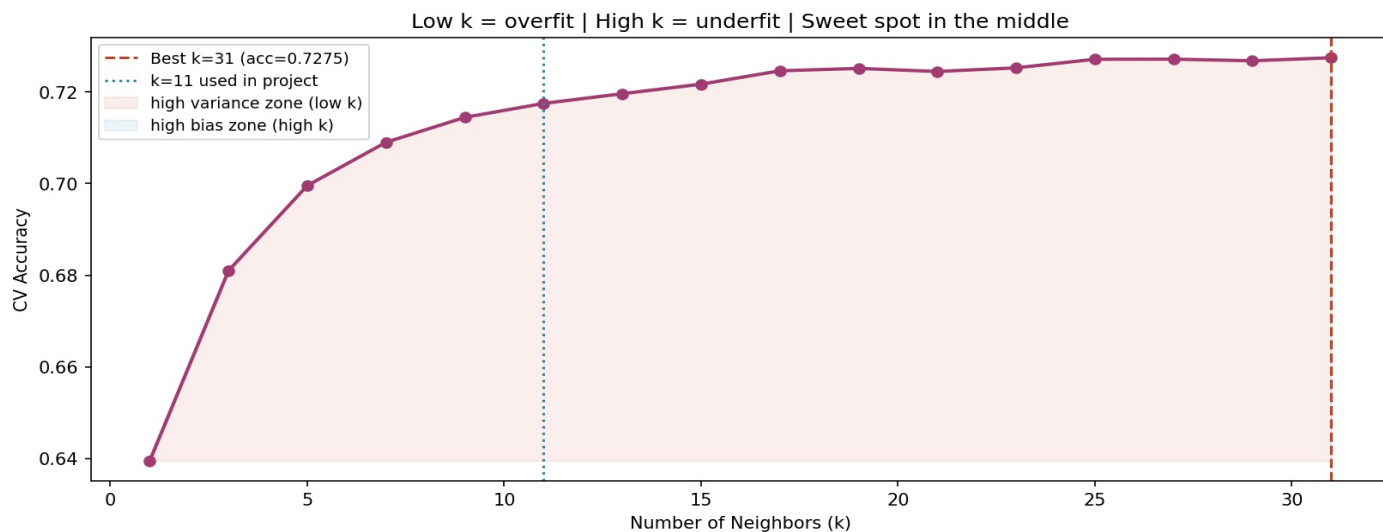


Figure 17. KNN optimal k search. Accuracy keeps rising through k=31, suggesting the data does not have strong local clustering. CVD and non-CVD patients are intermixed in feature space.

Fig 18 - Feature Importance Analysis

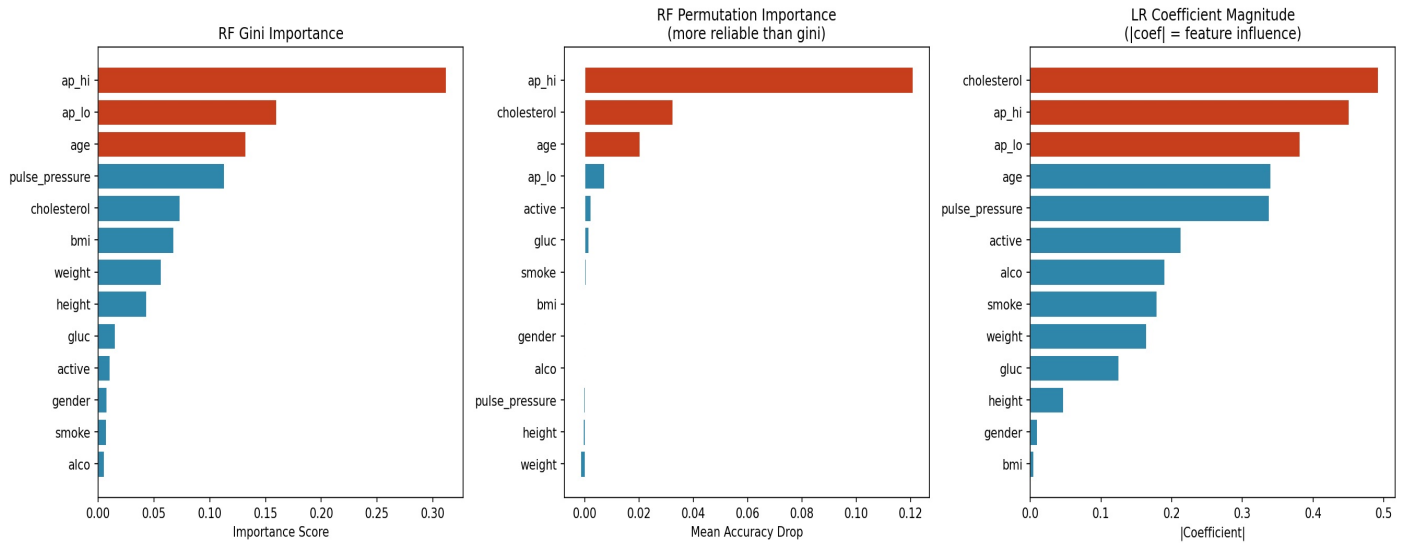


Figure 18. Feature importance comparison: RF gini, RF permutation, and LR coefficients. Permutation importance is the most reliable method. *ap_hi*, cholesterol, and age are consistently the top 3 across all three methods.

Fig 19 - Decision Tree (top 3 levels, acc=0.726)

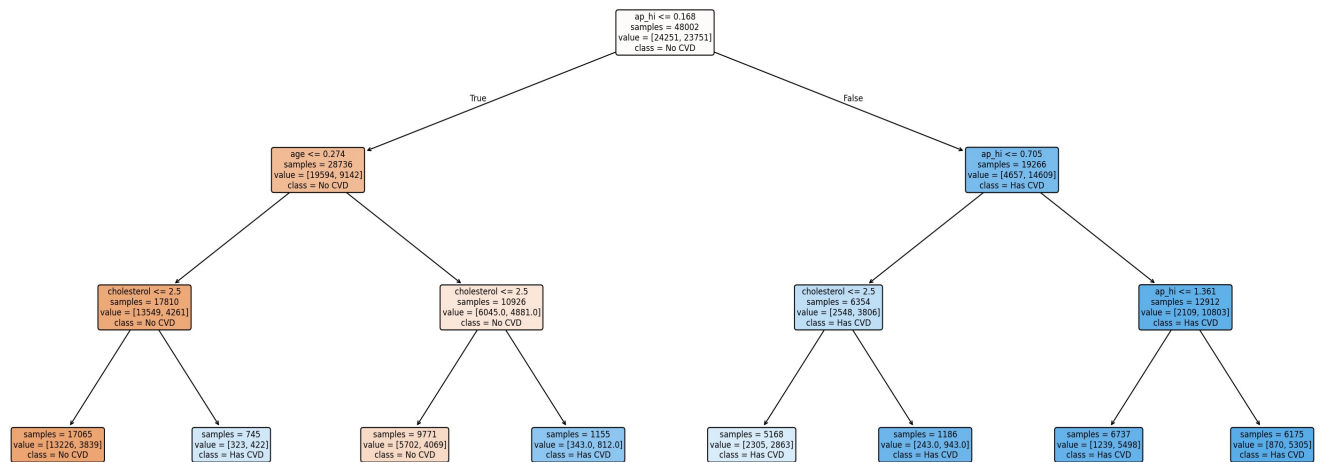


Figure 19. Decision tree visualization (top 3 levels). The root node splits on *ap_hi* at a threshold corresponding to approximately 130 mmHg in raw units, which is the AHA hypertension boundary. The model independently rediscovered this clinical threshold from data.

8. Cost-Sensitive Clinical Evaluation

This section is the core contribution of the project. We assign real-world costs to each type of mistake and rerank all 6 models by total clinical cost instead of accuracy.

Cost Assumptions

False Negative cost: \$150,000. This represents the total economic burden of a missed CVD event including emergency hospitalization, possible bypass surgery, cardiac rehabilitation, medications, and lost productivity. This is an estimate based on published healthcare economics literature and represents an upper-bound scenario.

False Positive cost: \$300. This represents the cost of an unnecessary outpatient cardiology referral, based on published consultation rates of \$150 to \$500.

The 500:1 ratio is what drives the entire analysis. Accuracy treats both errors equally. The cost model does not.

Results Per 10,000 Patients

Model	Accuracy	FN per 10K	FP per 10K	Total Cost	Acc Rank	Cost Rank
Random Forest	0.734	1,563	1,096	\$234.8M	#1	#2
KNN	0.719	1,498	1,309	\$225.1M	#5	#1
Decision Tree	0.730	1,577	1,125	\$236.9M	#3	#3
Logistic Regression	0.729	1,622	1,088	\$243.6M	#4	#4
SVM (RBF)	0.730	1,631	1,068	\$244.9M	#2	#5
Naive Bayes	0.716	1,913	925	\$287.2M	#6	#6

Orange = accuracy winner (RF). Green = clinical cost winner (KNN). Switching from RF to KNN saves \$9.71M per 10,000 patients and prevents 65 additional missed CVD diagnoses.

Why KNN Wins on Cost Despite Lower Accuracy

KNN misses 1,498 CVD patients per 10,000 screened. Random Forest misses 1,563. That difference of 65 patients per 10,000 translates to $65 \times \$150,000 = \9.75M in avoided emergency costs. KNN achieves this by making more false positives (unnecessary referrals) instead, which cost only \$300 each. The net result is a lower total cost even though KNN's overall accuracy is 1.5 percentage points lower.

This is not a cost cutting argument. It is a lives argument. KNN sends fewer sick patients home undiagnosed. That is what makes it the safer clinical choice.

Fig 20 - Cost-Sensitive Clinical Evaluation

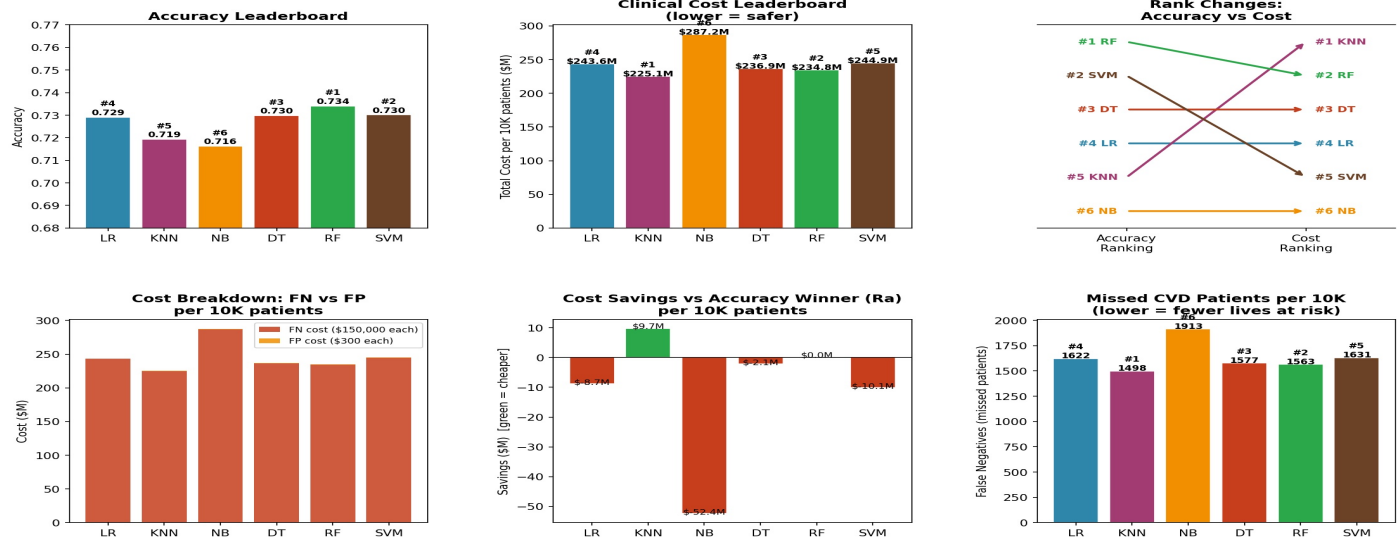


Figure 20. Cost-sensitive clinical evaluation. Top row: accuracy ranking, cost ranking, and rank change arrows showing the flip. Bottom row: FN vs FP cost breakdown, savings vs accuracy winner, and missed patients per 10K.

Sensitivity Analysis

The \$150,000 FN cost is an estimate. We tested whether the conclusion holds at different cost assumptions.

FN Cost Assumption	Cost Winner	Winner Cost per 10K
\$25,000	KNN	\$37.8M
\$50,000	KNN	\$75.3M
\$75,000	KNN	\$112.8M
\$100,000	KNN	\$150.2M
\$125,000	KNN	\$187.7M
\$150,000	KNN	\$225.1M
\$200,000	KNN	\$300.0M

KNN wins at every FN cost assumption from \$25,000 to \$200,000. The conclusion is robust to the exact cost figures chosen.

Fig 21 - Sensitivity Analysis: Does the Winner Change?

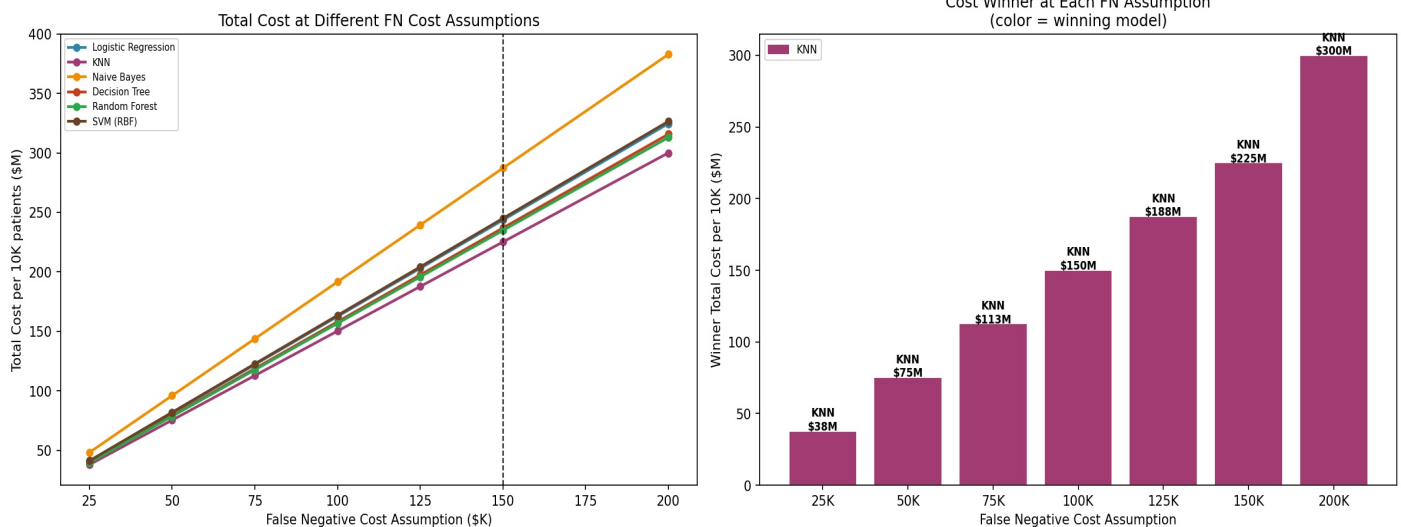


Figure 21. Sensitivity analysis showing total cost lines across FN cost assumptions (left) and cost winner at each assumption level (right). All bars are the same color because KNN wins at every level tested.

9. Minimal Feature Discovery

We asked how many features a doctor actually needs to achieve acceptable accuracy. Using permutation importance to order features by predictive contribution, we added one feature at a time and tracked accuracy for Random Forest, Logistic Regression, and KNN.

The permutation importance ranking was: ap_hi, cholesterol, age, ap_lo, active, gluc, smoke, bmi, gender, alco, pulse_pressure, height, weight.

Features	Feature Set	RF Accuracy	Gap from Full Model
1	ap_hi	0.714	-0.020
2	ap_hi + cholesterol	0.726	-0.008
3	ap_hi + cholesterol + age	0.730	-0.004
4	+ ap_lo	0.731	-0.003
5	+ active	0.734	0.000
13	All features	0.734	0.000

The accuracy curve plateaus at 3 features. The best 3-feature combination (ap_hi, cholesterol, age) achieves 0.730 accuracy versus 0.734 for all 13 features. The gap is 0.4%, which is clinically negligible.

Clinical meaning: a primary care doctor who has only blood pressure, cholesterol level, and the patient's age has almost everything the model needs. No additional lab work or questionnaire is required to get 99.5% of full model accuracy.

Fig 22 - Minimal Feature Discovery

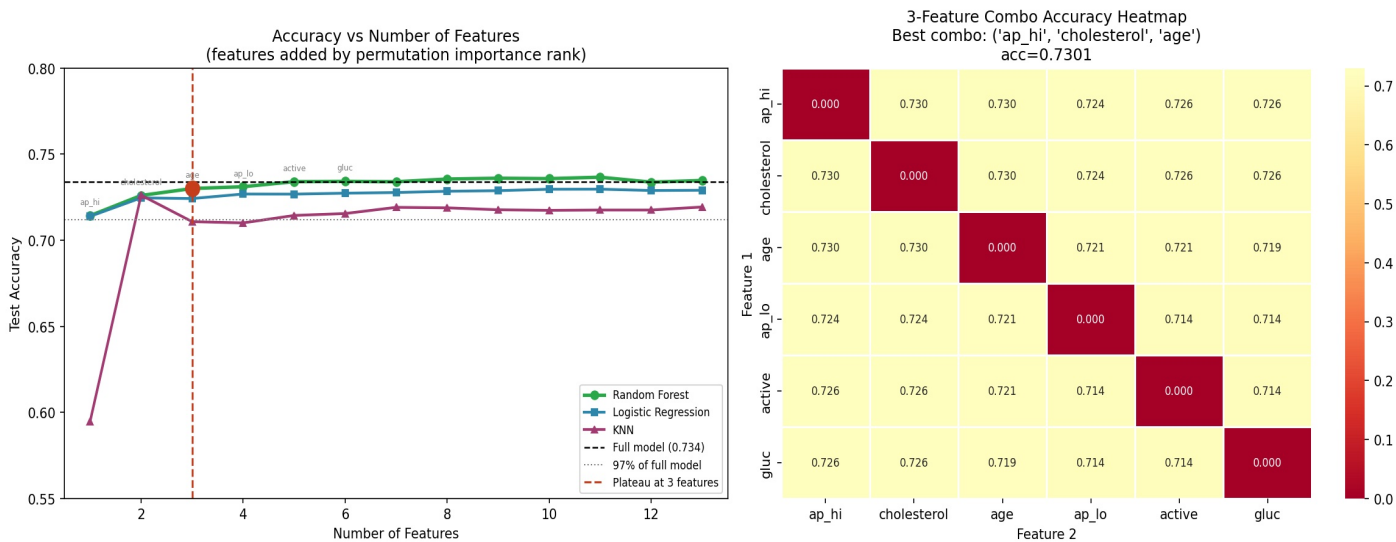


Figure 22. Accuracy vs number of features (left) and 3-feature combination heatmap (right). The plateau at 3 features is clear. The heatmap shows ap_hi and cholesterol consistently produce the best accuracy regardless of what third feature is added.

10. Fairness, Robustness, and Confidence Analysis

10.1 Gender Fairness Audit

The dataset is 65.1% female (44,659 patients) and 34.9% male (23,916 patients). This imbalance could cause models to learn female patterns better than male patterns, creating a recall disparity where CVD is more likely to be missed in male patients.

We measured the absolute difference in recall between genders for each model. A gap above 5% is considered a fairness concern in clinical AI literature.

Model	Recall Female	Recall Male	Recall Gap	Pass?
Random Forest	0.685	0.682	0.003	YES
KNN	0.700	0.693	0.007	YES
Naive Bayes	0.615	0.610	0.006	YES
Decision Tree	0.677	0.689	0.012	YES
Logistic Regression	0.668	0.680	0.011	YES
SVM (RBF)	0.666	0.678	0.012	YES

All 6 models pass. The maximum recall gap is 1.2% (Decision Tree and SVM), well below the 5% threshold. We also ran an identical patient pair test: a 55-year-old patient with BP 140/90 and above-normal cholesterol was presented to all 6 models twice, once as female and once as male. Every model gave the same prediction with probability differences under 0.01. No model introduces gender bias in individual predictions.

Fig 23 - Gender Fairness Audit

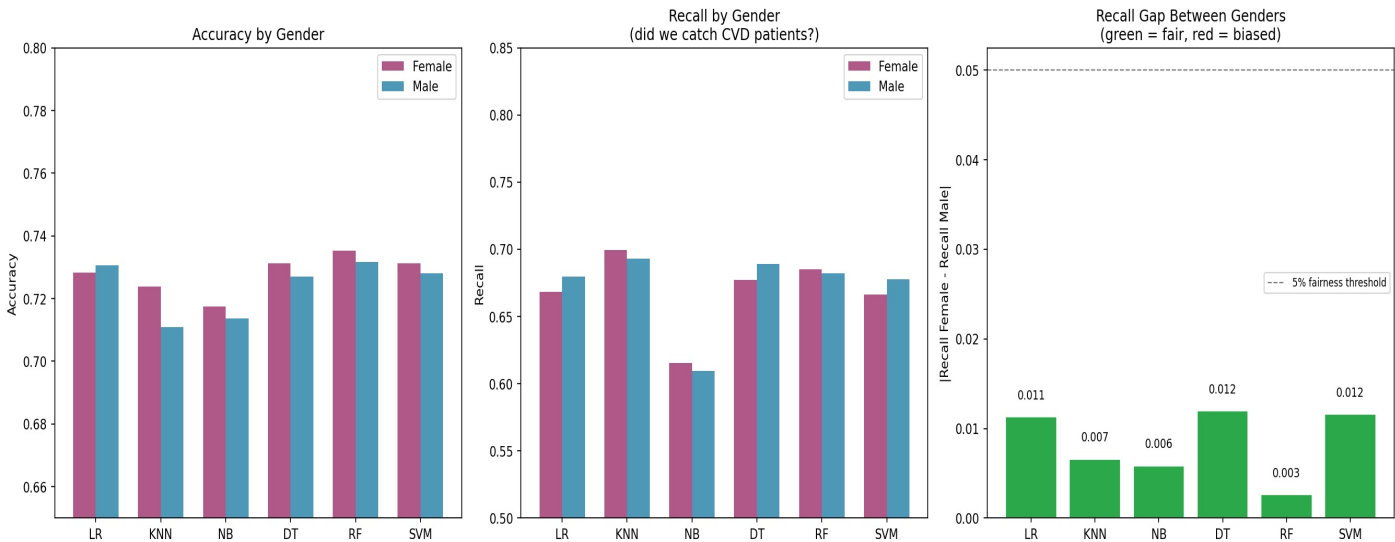


Figure 23. Gender fairness audit: accuracy by gender, recall by gender, and recall gap. All bars are green, all below the 5% threshold.

10.2 Robustness Under Measurement Noise

Real hospital measurements are not perfect. Blood pressure readings fluctuate several mmHg between measurements. We simulated this by adding Gaussian noise at 1%, 5%, 10%, and 20% levels to the scaled test features and re-evaluated all 6 models.

Model	Clean Accuracy	Accuracy at 20% Noise	Drop	Robust?
Logistic Regression	0.729	0.728	0.001	YES
KNN	0.719	0.719	0.000	YES
Naive Bayes	0.716	0.717	-0.001	YES
Decision Tree	0.730	0.698	0.032	NO
Random Forest	0.734	0.731	0.003	YES
SVM (RBF)	0.730	0.729	0.001	YES

Five of six models are robust. Decision Tree fails with a 3.2% accuracy drop at 20% noise. This is expected: a single tree makes hard splits at fixed thresholds, and measurement noise frequently pushes feature values across those thresholds, changing the prediction. Random Forest, which averages 100 trees, is almost immune to this effect (drop of 0.003).

Fig 24 - Robustness Under Measurement Noise

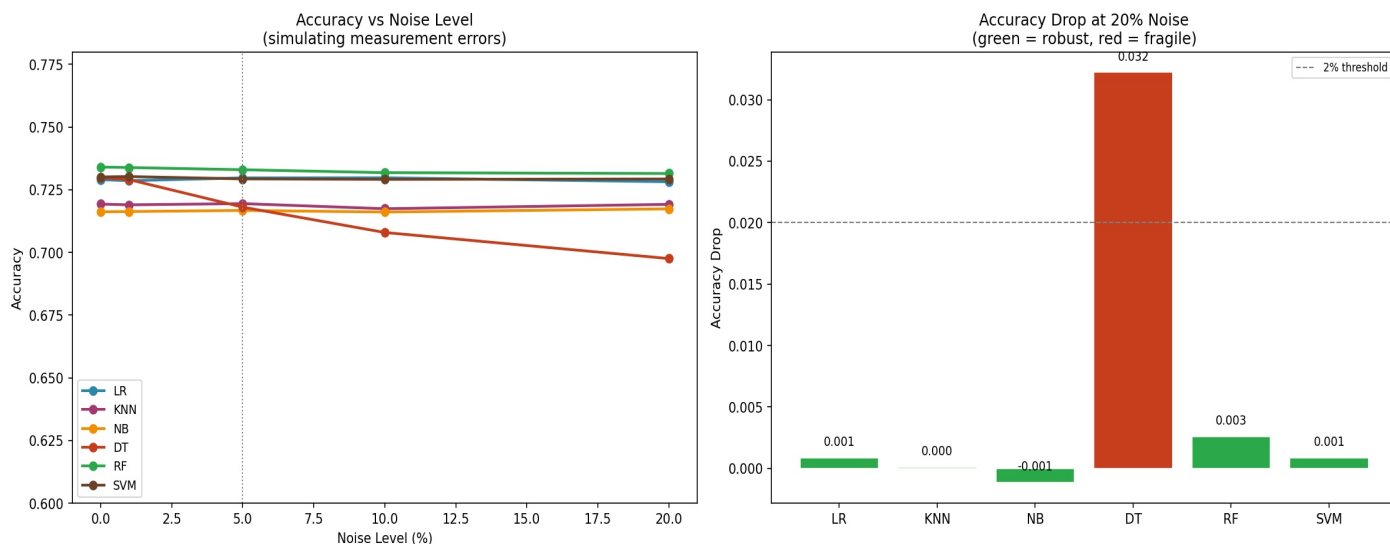


Figure 24. Robustness analysis. Left: accuracy vs noise level for all 6 models. Right: accuracy drop at 20% noise. Decision Tree (red bar) is the only model that fails the 2% threshold.

10.3 Confidence-Based Predictions

Instead of always forcing a binary prediction, we implemented a grey zone. When the Random Forest outputs a probability between 0.40 and 0.60, the model reports 'uncertain' rather than guessing. These patients are flagged for additional testing.

Results for Random Forest with a 0.40 to 0.60 grey zone:

Confident predictions: 16,678 out of 20,573 test patients (81.1%)

Grey zone (uncertain): 3,895 patients (18.9%)

Overall accuracy on all patients: 73.4%

Accuracy on confident predictions only: 77.9%

Gain from filtering uncertain cases: +4.5 percentage points

A model that says 'I do not know' on 19% of cases is more clinically trustworthy than one that always guesses. The coverage vs accuracy tradeoff curve shows that as the confidence threshold is tightened, accuracy can reach 91% on the most confident predictions at the cost of covering fewer patients.

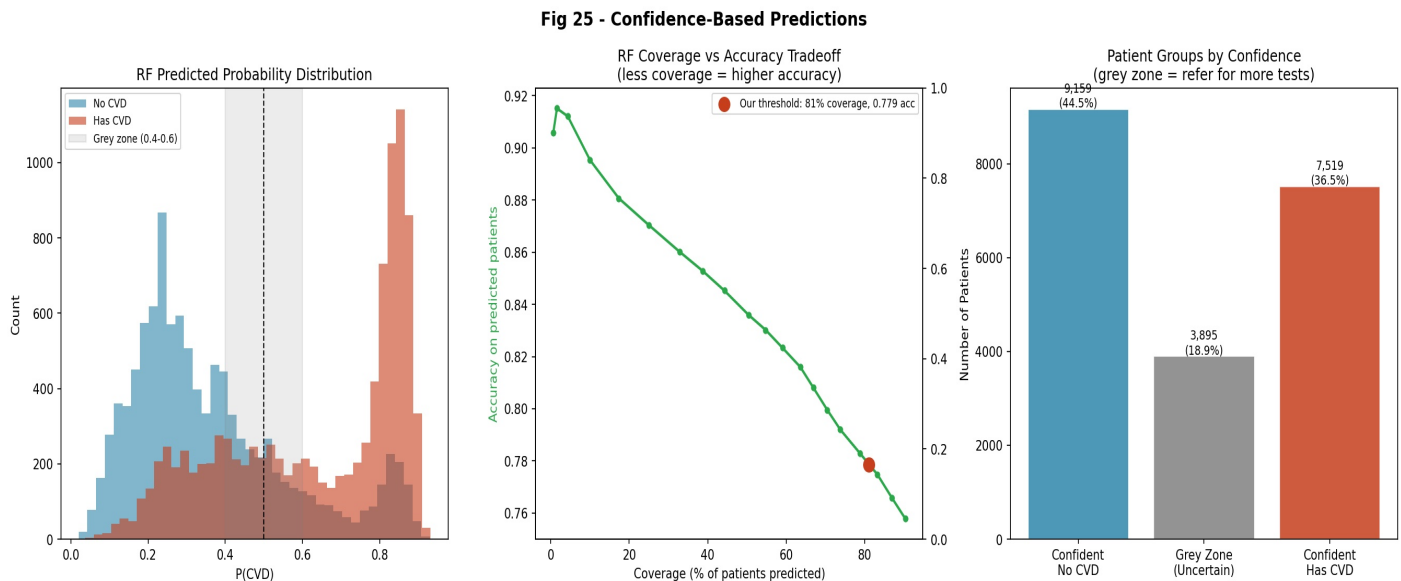


Figure 25. Confidence analysis: probability distribution with grey zone, coverage vs accuracy tradeoff, and patient group breakdown.

11. Clinical Risk Calculator

We built an interactive risk calculator using the trained Random Forest model with confidence filtering. A doctor types in a patient's age, blood pressure, and cholesterol level and receives a predicted CVD probability, a risk level (Low/Uncertain/High), and a recommended action.

Demo Cases

Patient Profile	P(CVD)	Risk Level	Action
Age 38, BP 110/70, Normal Chol, BMI 24.1	13.5%	LOW RISK	No referral needed
Age 52, BP 135/85, Above Normal Chol, BMI 27.1	64.4%	HIGH RISK	Refer to cardiologist
Age 62, BP 165/100, Well Above Chol, BMI 35.2	84.9%	HIGH RISK	Refer immediately
Age 50, BP 130/85, Above Normal Chol, BMI 26.4	51.3%	UNCERTAIN	Order more tests
Age 60, BP 150/95, Well Above Chol, BMI 34.0	85.0%	HIGH RISK	Refer to cardiologist

Case 4 (yellow) lands in the grey zone at 51.3% because it sits right at the clinical boundary. The model correctly identifies this as uncertain rather than forcing a potentially wrong binary decision.

12. Conclusions and Limitations

Key Findings

Accuracy winner: Random Forest at 73.4% accuracy and AUC 0.801

Clinical cost winner: KNN at \$225.1M per 10,000 patients (vs RF at \$234.8M)

Savings from switching RF to KNN: \$9.71M per 10,000 patients

Fewer missed CVD patients with KNN: 65 fewer per 10,000 screened

Minimal features needed: 3 features (ap_hi, cholesterol, age) = 99.5% of full model accuracy

Gender fairness: All 6 models pass with max recall gap of 1.2%

Robustness: 5 of 6 models hold under 20% noise. Decision Tree fails (3.2% drop)

Confidence filtering: RF achieves 77.9% accuracy on confident predictions (81% of patients)

Sensitivity proof: KNN wins at every FN cost assumption from \$25,000 to \$200,000

Clinical Recommendation

For a primary care CVD screening tool, KNN should be preferred over Random Forest because it makes fewer false negatives. Fewer false negatives means fewer sick patients sent home undiagnosed. For applications where AUC and overall discrimination matter more (such as research scoring), Random Forest is the better choice. Decision Tree should not be deployed in production settings due to its sensitivity to measurement noise.

The recommended production setup is Random Forest with 0.40 to 0.60 confidence filtering. Patients in the grey zone should be referred for additional testing rather than receiving a forced binary prediction.

Limitations

No laboratory biomarkers: The dataset contains only physical measurements and self-reported lifestyle data. Real cardiology diagnosis incorporates ECG readings, troponin levels, and LDL/HDL cholesterol fractions. Adding these would likely push accuracy from the current 73-74% ceiling toward 85-90%.

Self-reported lifestyle data: Smoking, alcohol use, and physical activity are self-reported. Patients commonly underreport these behaviors in medical settings. This explains why smoke, alco, and active showed near-zero signal in permutation importance analysis despite being known CVD risk factors in clinical literature.

No external validation: The model was trained and tested on one dataset. We do not know whether it generalizes to patients from a different country, hospital system, or demographic population. External validation on an independent clinical dataset is required before any real-world deployment.

Cross-sectional data: Each patient appears once. We do not know if they developed CVD 6 months or 10 years after their measurement. A longitudinal design would allow predicting future CVD onset, which is more clinically useful than predicting current status.

Cost estimates: The \$150,000 and \$300 figures are evidence-based estimates and not exact values. They vary by insurance system, hospital, and country. The sensitivity analysis shows the conclusion holds from \$25,000 to \$200,000, but a real deployment would need cost data from the specific healthcare system.

SVM subsample: SVM was trained on 10,000 rows due to computational constraints. Full-data SVM results may differ slightly.

Future Work

Add ECG and laboratory biomarker data to push accuracy above 80%

Validate on independent clinical datasets such as UK Biobank or NHANES

Implement threshold optimization per demographic subgroup for improved equity

Run a prospective study to measure whether model-guided referrals improve patient outcomes versus unassisted physician judgment

Regulatory pathway analysis for FDA clinical decision support classification

Calibration analysis to ensure probability outputs are reliable for clinical use

13. Technical Appendix

Environment and Libraries

Python 3.13 on macOS. All computation runs locally.

Library	Version	Purpose
pandas	latest	Data loading, cleaning, manipulation
numpy	latest	Array operations, random seeding
scikit-learn	latest	All 6 classifiers, scaling, CV, metrics
matplotlib	latest	All figures
seaborn	latest	Heatmap and pairplot
reportlab	latest	This report

Reproducibility

All experiments use `RANDOM_STATE = 42`. The same seed is passed to `train_test_split`, `StratifiedKFold`, `RandomForestClassifier`, `DecisionTreeClassifier`, `LogisticRegression`, and `SVC`. Results are fully reproducible.

Cross-Validation Setup

`StratifiedKFold` with `n_splits=10`, `shuffle=True`, `random_state=42`. All 6 models used identical folds, making their CV scores directly comparable. SVM CV was run on the 10,000-row subsample.

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